In [ ]:

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# PORTFOLIO #2: CONDITIONALS
# CONCEPT: SPECIATION ANALYZER
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In an analytical chemistry calculation, the state of an acid (protonated/deprotonated) can be determined based on the relationship between the pH of the solution and the pKa of the chemical substance being analyzed.

This code uses conditional statements to determine if the state of an acid is either in acidic or basic form, and if it is protonated or deprotonated.

In [19]:

```
pH = 3.75 # Assigns a value to pH
pKa = 4 # Assigns a value to pKa

print("current pH:", pH) # Shows the given pH in the result
print("substance pKa:", pKa) # Shows the given pKa in the result
print ("-" * 30) # Divides the given to the final answer
x = pH - pKa # Assigns a variable for easier access


if x <= -1:
    # pH is at least 1 unit below pKa (mostly protonated)
    print('Dominant Species: Acidic Form [HA]. Protonated.')

elif x >= 1:
    # pH is at least 1 unit above pKa (mostly deprotonated)
    print('Dominant Species: Basic Form [A-]. Deprotonated.')

else:
    # pH is near the pKa (buffer region)
    percentage_A = (10**x / (1 + 10**x)) * 100
    print('Buffer Region. Mixture of forms. [A-] is approximately', percentage_A,"%
```

```
current pH: 3.75
substance pKa: 4
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Buffer Region. Mixture of forms. [A-] is approximately 35.9935000197115 %
```